Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

The Model Substitution Problem:

Economic Incentives Drive Covert Swaps

Core Problem

Commercial LLM API providers may substitute advertised models (e.g., Llama-3-70B) with cheaper alternatives to reduce GPU costs while maintaining pricing.

Four Substitution Types

1. **Smaller Model**
   e.g., 8B instead of 70B
2. **Quantized Variant**
   FP8/INT8 instead of FP16
3. **Fine-tuned Version**
   Modified weights
4. **Different Family**
   e.g., Mistral instead of Llama

Formal Hypothesis Testing

H_0 (Null Hypothesis)

API serves specified model

P_spec

↓

H_1 (Alternative Hypothesis)

API serves substitute model

P_actual

Auditor's Challenge

Distinguish $P\_spec(y|x)$ from $P\_actual(y|x)$

Response distributions differ based on model architecture and weights

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

For most models, FP8 scores fall within error bars of original — detection is extremely challenging.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Log probability variation across inference frameworks and hardware for Llama-3-8B. Source: arxiv.org/abs/2504.04715

Key Findings

Same Llama-3-8B model produces different log probabilities across:

- Software frameworks: transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2
- GPU hardware: A100 vs H100

This production-level nondeterminism defeats metadata-based detection

Auditor cannot distinguish legitimate model variation from actual substitution

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Key Insight: Detection requires exponentially more queries as p decreases

```
P_mixed(y|x) = (1-p) · P_spec(y|x) + p · P_actual(y|x)
```

Attack 1: Randomized Substitution

Provider mixes substitute responses with probability p

Mixed distribution:

1. `(1-p) × P_spec + p × P_substitute`

As p → 0, mixed distribution converges to specified model's distribution

Attack 2: Benchmark Evasion

Provider detects known benchmark prompts

Routes them to the genuine model

Serves substitutes for regular traffic

Auditor sees high benchmark performance but degraded actual capability

Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Weakness	Practical Limitation
Text-based statistical tests (e.g., MMD)	Requires large query budgets; fails against subtle or randomized substitutions	Impractically high detection threshold
Log-probability methods	Defeated by production inference nondeterminism across software versions and GPU hardware	Cannot distinguish legitimate variation from substitution
Adversarial countermeasures (benchmark evasion, randomized routing)	Undermines all detection approaches	Makes statistical detection query-prohibitive

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

How TEEs Work

- ✓ Hardware-secured enclaves provide cryptographic attestation
- ✓ Prove that specific model weights are loaded
- ✓ Prove that correct inference code is running
- ✓ Only modest performance overhead

Software Methods vs TEEs

Property	Software Methods	TEEs
Provable	No	**Yes**
Efficient	Yes (but unreliable)	Yes
Robust	No	Yes

Practical Deployment

NVIDIA confidential computing on H100 GPUs

Unlike ZKPs

Computationally prohibitive for large models

TEEs are practical and deployable today

Key Takeaways: Close the Verification Gap with Hardware Attestation

1. Model substitution is economically motivated and practically undetectable via software
2. FP8 quantization barely affects benchmarks — making substitution nearly invisible
3. Inference nondeterminism kills log-probability fingerprinting in production
4. Randomized substitution defeats statistical tests — detection becomes query-prohibitive
5. TEEs are the only currently viable path to verifiable LLM APIs

The research community and cloud infrastructure operators must adopt hardware attestation to close the verification gap.